INCH-POUND
MIL-DTL-24643/45G
27 November 2023
SUPERSEDING
MIL-DTL-24643/45F(SH)
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/45F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LS2U (INCLUDING VARIATION LS2WU)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS2U (non-watertight) and LS2WU (watertight).

Construction (Non-Watertight and Watertight)

First	-	Copper conductor, tin-coated, ASTM B286, 26 AWG with 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.015 inch. Wall thickness may vary from nominal as necessary in order for completed cable to meet the specified electrical requirements. Telephone identification code shall be applied by method 3.
Fourth	-	Two conductors cabled together with a maximum lay of 1½ inches to form a pair.
Fifth	-	The specified number of pairs (see table I or table II) shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Braided shield of tin-coated copper.
Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. See table I or table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details (non-watertight).

Military part no. M24643/45	Type and size	No. of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inch)	
				min.	max.
-01UO	LS2U-10	10	0.050	0.450	0.480
-02UO	LS2U-15	15	0.050	0.530	0.560
-03UO	LS2U-19	19	0.050	0.550	0.580
-04UO	LS2U-30	30	0.050	0.670	0.700
-05UO	LS2U-45	45	0.050	0.830	0.870
-06UO	LS2U-60	60	0.065	0.920	0.960

TABLE II. Details (watertight).

Military part no. M24643/45	Type and size	No. of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inch)	
				min.	max.
-07UO	LS2WU-10	10	0.050	0.460	0.490
-08UO	LS2WU-15	15	0.050	0.540	0.570
-09UO	LS2WU-19	19	0.050	0.560	0.590
-10UO	LS2WU-30	30	0.050	0.690	0.720
-11UO	LS2WU-45	45	0.050	0.850	0.890
-12UO	LS2WU-60	60	0.065	0.940	0.980

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	45.19
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to overall shield	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

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Group A:	
Visual and dimensional	No failure
Watertightness (type LS2WU only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Characteristic impedance at 1 megahertz (ohms)	135±15
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (type LS2WU only)	No failure
Bending endurance at 77±3.6 °F (25±2 °C) (cycles, min.)	25
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

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Cable fill (sealant) removability (type LS2WU only)	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure

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Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2U (all sizes)	1,000
LS2WU (all sizes)	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-061)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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